

Covariance Estimation under Portfolio Constraints in Mexican Equities: Static and Rolling Evidence

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Abstract

Does covariance estimation materially change minimum-variance portfolio outcomes in Mexican equities once long-only constraints and periodic rebalancing are considered? We compare sample, eigenvalue-clipped, Ledoit–Wolf, and rotationally invariant (RIE) covariance estimators in a 37-stock Bolsa Mexicana de Valores panel. Daily gross total returns provide 3,706 observations; portfolio comparisons use simple returns, a common chronological holdout, long-only portfolios, an equally weighted benchmark, and a 504-observation rolling window with 21-observation buy-and-hold periods. In the fixed holdout, unconstrained RIE reduces daily volatility by 0.0046 percentage points relative to the sample covariance estimator, but its annualized Sharpe difference is -0.025 (20-day moving-block 95% interval $[-0.045, -0.004]$; HAC $p = 0.015$). Under long-only constraints, RIE’s Sharpe is -0.035 , compared with 0.450 for $1/N$. In the rolling design, long-only $1/N$ attains a Sharpe of 0.767 , while RIE attains -0.646 ; RIE’s Sharpe difference relative to $1/N$ is -1.413 (21-day block-bootstrap 95% interval $[-2.333, -0.641]$). Hypothetical proportional-cost scenarios do not reverse this ranking, but no transaction costs are directly observed. The results do not establish a generalizable estimator ranking. They show that investor-relevant constraints, rebalancing, and comparison with a naive benchmark materially change the assessment of covariance cleaning in this selected Mexican-equity panel.

Keywords: emerging markets; Mexican equities; portfolio selection; covariance estimation; minimum-variance portfolio; transaction costs; out-of-sample evaluation.

1 Introduction

Covariance estimates are central to minimum-variance portfolio construction: the optimizer inverts the estimated risk matrix, so sampling error in small eigenvalues can lead to large and unstable positions. Spectral cleaning and shrinkage regularize this input, but a lower estimated portfolio variance does not ensure superior realized risk-adjusted returns or a feasible investment strategy. The comparison is especially relevant in emerging equity markets, where investable universes can be concentrated and trading frictions and data coverage can be important to portfolio design [6].

The empirical question is *how sensitive are minimum-variance portfolio outcomes to covariance estimation when Mexican equities are evaluated under long-only constraints and periodic rebalancing?* We compare four estimators on one common daily total-return panel, include the equally weighted $1/N$ portfolio, and distinguish a static-weight holdout from a rolling buy-and-hold protocol. The application is a single-country analysis. It speaks to the choice of performance benchmarks

and the limitations of covariance-based allocation in a Mexican equity sample; it does not estimate a regional emerging-market effect or a causal institutional mechanism.

Three results motivate this distinction. First, in the fixed holdout, RIE has lower realized volatility than the sample covariance estimator but a lower Sharpe ratio. Second, the simple $1/N$ portfolio has a substantially higher point estimate of Sharpe than the long-only covariance-based portfolios. Third, monthly rolling comparisons preserve the large point estimate gap between $1/N$ and RIE, while estimated turnover and assumed cost scenarios are reported explicitly. The analysis is deliberately conditional on a retrospectively assembled 37-stock universe and one historical path.

2 Data and methods

2.1 Daily total-return panel

The panel contains 37 Mexican equities mapped one-to-one from the original 38-security candidate list to issuer tabs in the Bloomberg workbook downloaded in September 2026. We use Bloomberg’s `TOT_RETURN_INDEX_GROSS_DVDS` field rather than unadjusted `PX_LAST` prices. The 37 mapped tickers are MFRISCOA-1, RA, SIMECB, ICHB, GENTERA, SORIANAB, ELEKTRA, CHDRAUIB, LIVEPOLC-1, AUTLANB, HERDEZ, GAPB, OMAB, MEGACPO, PINFRA, LABB, ALSEA, ARA, ASURB, AMXB, AC, AXTELCPO, KIMBERA, CEMEXCPO, BIMBOA, BOLSA, GCARSOA1, FEMSAUBD, GFNORTEO, GFINBURO, ORBIA, GRUMAB, GMEXICOB, PE&OLES, TLEVISACPO, SIGMAFA, and WALMEX. FUNO11 was absent from the workbook and was excluded rather than replaced. The balanced panel runs from 18 July 2011 through 29 September 2025 and contains 3,706 daily log returns with no gaps longer than seven calendar days.

The last 20% of observations (741 dates, 28 November 2022–29 September 2025) form a chronological holdout. Covariance matrices are estimated on the preceding 2,965 dates, so $q = N/T_{\text{train}} = 37/2965 \approx 0.0125$. The security set is retrospective, not reconstructed from historical BMV/BIVA membership. Its balanced coverage therefore does not remove survivorship, ticker-mapping, or ex-post selection bias. The 37 names are not treated as a representative index or as a complete investable universe.

2.2 Covariance estimators and portfolios

We compare the sample covariance (SM), Marčenko–Pastur eigenvalue clipping (CL), Ledoit–Wolf identity-target linear shrinkage (LW), and analytic rotationally invariant nonlinear shrinkage (RIE). CL standardizes the covariance to a correlation matrix, retains eigenvalues above the Marčenko–Pastur upper edge, replaces the remaining bulk eigenvalues by their average, and restores the original marginal scales. RIE applies kernel-based analytic shrinkage to sample eigenvalues while retaining sample eigenvectors [2, 3, 4]. The implementation is designed for $q < 1$.

For each covariance estimate $\hat{\Sigma}$, the unconstrained global minimum-variance weights are

$$\hat{w} = \frac{\hat{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \hat{\Sigma}^{-1}\mathbf{1}}, \quad \mathbf{1}^\top \hat{w} = 1.$$

The long-only version minimizes $w^\top \hat{\Sigma} w$ subject to $\mathbf{1}^\top w = 1$ and $0 \leq w_i \leq 1$. We also evaluate the equally weighted portfolio $w_i = 1/N$, a standard benchmark in portfolio selection [5].

For each asset, $\ell_{i,t}$ denotes the log change in its gross total-return index; the simple return is $r_{i,t} = \exp(\ell_{i,t}) - 1$. In the static holdout, covariance estimates and target weights are fit once on

the training sample, and the portfolio is restored to its target weights daily: $r_{p,t} = \sum_i w_i r_{i,t}$. This convention is distinct from holding shares fixed throughout the test period.

The rolling sensitivity estimates each covariance matrix from the preceding 504 observations, sets weights every 21 observations, and holds shares fixed within each 21-observation block. Portfolio wealth is reconstructed from asset-price relatives within each block. Twenty-five non-overlapping blocks cover 26 September 2023–29 September 2025. This rolling analysis shares its test observations with the fixed holdout and is a protocol sensitivity, not an independent replication.

2.3 Performance, turnover, and inference

We report annualized arithmetic mean return $252\bar{r}_p$, annualized volatility $\sqrt{252}s_p$, and annualized Sharpe $\sqrt{252}(\bar{r}_p - r_f/252)/s_p$, using a fixed 8% annual risk-free rate. Turnover is one-way traded notional divided by pre-trade portfolio value. For the static protocol, daily turnover measures the trade needed to restore target weights after market drift; in the rolling design, turnover is measured at each rebalance.

No transaction-cost observations are available. We therefore subtract hypothetical proportional costs of 0, 5, 10, 25, and 50 basis points per unit of one-way traded notional. Initial deployment costs, taxes, market impact, liquidity limits, and security-specific bid–ask spreads are not estimated. The cost scenarios are sensitivities, not measurements of Mexican implementation costs.

Paired moving-block bootstrap intervals use 20,000 replications and common resampled date indices (10-, 20-, and 42-observation blocks for the static holdout; 10-, 21-, and 42-observation blocks for the rolling comparison). HAC standard errors with 20 lags provide two-sided tests for Sharpe and volatility differences. Inference is conditional on the fitted weights and chosen universe and does not propagate covariance-estimation, universe selection, or model-selection uncertainty. The fixed-holdout RIE–SM comparison is the primary inferential contrast; the rolling and $1/N$ comparisons are sensitivity analyses. We do not adjust for the number of secondary comparisons.

3 Results

3.1 Fixed chronological holdout

Table 1 reports the four unconstrained portfolios. RIE has lower holdout volatility than SM (0.6952% versus 0.6998% per day), but its Sharpe is lower (−0.232 versus −0.207). Ledoit–Wolf has both the lowest volatility and the highest Sharpe among covariance-based estimators. The ranking therefore depends on the loss function; a volatility reduction by itself is not evidence of improved investor outcomes.

Table 1: Unconstrained portfolios on the fixed holdout. Volatility is percent per trading day; Sharpe is annualized.

Estimator	Training vol.	Holdout vol.	Sharpe	NEA	$\exp(H)$
Sample (SM)	0.7093	0.6998	−0.207	10.58	25.37
Clipping (CL)	0.7079	0.7084	−0.339	9.50	27.10
Ledoit–Wolf (LW)	0.7071	0.6965	−0.206	10.91	25.57
RIE	0.7097	0.6952	−0.232	10.85	25.91

The RIE-minus-SM daily volatility difference is −0.00462 percentage points (20-observation block-bootstrap 95% interval [−0.00713, −0.00113]; HAC $p = 0.0036$). The Sharpe difference is −0.0248 (95% interval [−0.0451, −0.0038]; HAC $p = 0.015$). The Sharpe interval remains below

zero at block lengths 10 and 42. Under long-only constraints, RIE’s Sharpe difference from SM is -0.0113 (20-observation interval $[-0.0242, 0.0028]$; HAC $p = 0.086$), while its volatility difference remains negative. These intervals are conditional on weights fitted in the training sample.

RIE’s unconstrained effective number of assets is 10.85 compared with 10.58 for SM; Shannon effective breadth is 25.91 versus 25.37. Both measures are far below the 37 assets of the equal-weight portfolio. These statistics describe portfolio weights and should not be interpreted as market-wide investor diversity.

3.2 Long-only portfolios and the naive benchmark

The long-only results sharpen the applied comparison. On the fixed holdout, $1/N$ has a Sharpe of 0.450 and an annualized arithmetic mean return of 13.72%, compared with -0.035 and 7.62% for RIE. The $1/N$ portfolio also has higher volatility (12.72% versus 11.10%). The paired RIE-minus- $1/N$ Sharpe estimate is -0.485 , but its interval is wide ($[-1.068, 0.053]$; HAC $p = 0.099$); the point-estimate ranking is not a precise estimate of expected performance beyond this sample.

Table 2: Long-only portfolios on the fixed holdout. Returns and volatility are annualized percentages; turnover is mean one-way traded notional per day.

Estimator	Mean return	Volatility	Sharpe	NEA	Turnover
Sample	7.74	11.18	-0.023	13.72	0.52%
Clipping	7.65	11.20	-0.031	13.25	0.51%
Ledoit–Wolf	7.76	11.13	-0.022	14.04	0.52%
RIE	7.62	11.10	-0.035	14.03	0.52%
$1/N$	13.72	12.72	0.450	37.00	0.61%

3.3 Rolling deployment and cost sensitivity

The 25-block rolling comparison does not establish a performance advantage for RIE over SM. For long-only portfolios, annualized Sharpe is -0.646 for RIE and -0.736 for SM; their difference of $+0.090$ has a 21-observation block-bootstrap 95% interval $[-0.052, 0.185]$ and HAC $p = 0.176$. The long-only $1/N$ portfolio attains a Sharpe of 0.767 and a mean return of 18.19%. Its Sharpe difference from RIE is 1.413 (RIE minus $1/N$ is -1.413 ; 95% interval $[-2.333, -0.641]$; HAC $p = 0.0013$). This contrast is exploratory because it uses the same test history and one set of model choices.

Mean one-way turnover per actual rebalance is 5.05% for long-only RIE and 2.81% for $1/N$. At the assumed upper cost of 50 basis points per one-way turnover, annualized mean return is 0.57% for RIE and 18.03% for $1/N$; at 10 basis points, the corresponding estimates are 0.80% and 18.16%. These differences do not establish expected future returns: the assumed costs are not transaction data, and initial portfolio formation costs are excluded.

4 Discussion

The central empirical result is not that one covariance estimator dominates. In the fixed holdout, RIE achieves slightly lower volatility but a lower Sharpe than the sample estimator; Ledoit–Wolf performs better on both reported risk and Sharpe point estimates among the covariance-based unconstrained portfolios. Long-only estimation reduces reliance on short positions but does not make covariance optimization outperform the naive benchmark in this sample. The rolling analysis

The long-only 1/N benchmark has the highest holdout Sharpe

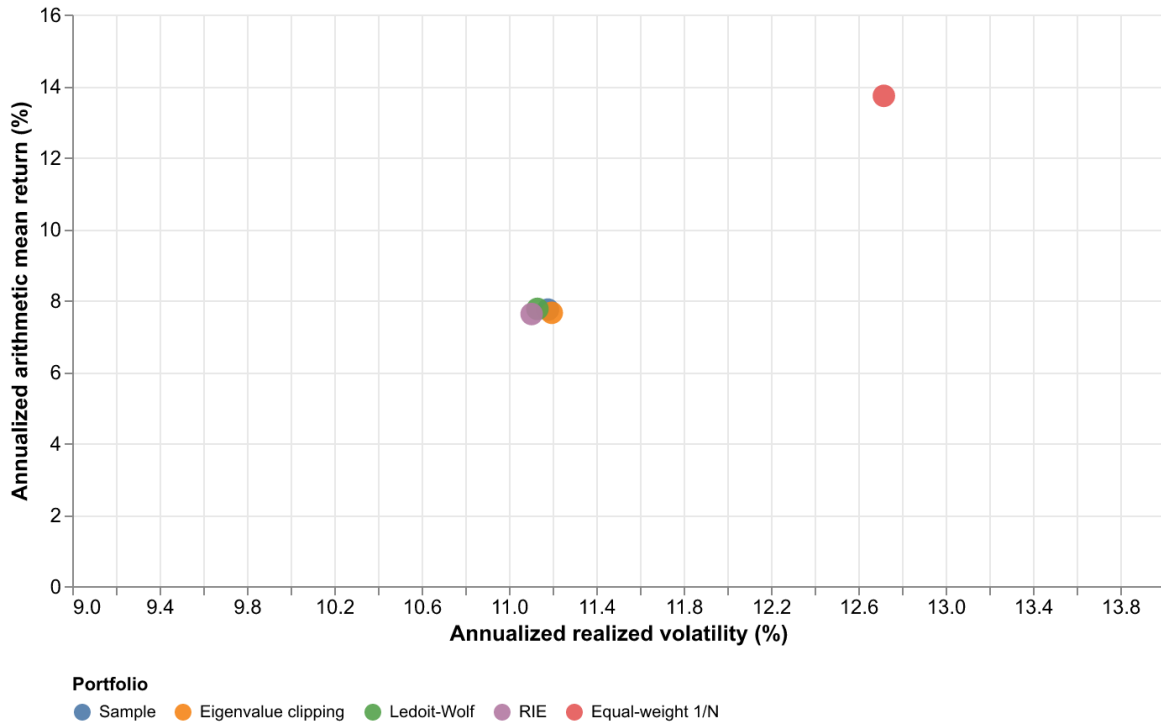


Figure 1: Annualized mean return and volatility for long-only portfolios on the fixed holdout. 1/N has the highest mean return and Sharpe, with higher realized volatility.

Rolling results are exploratory and depend on constraints

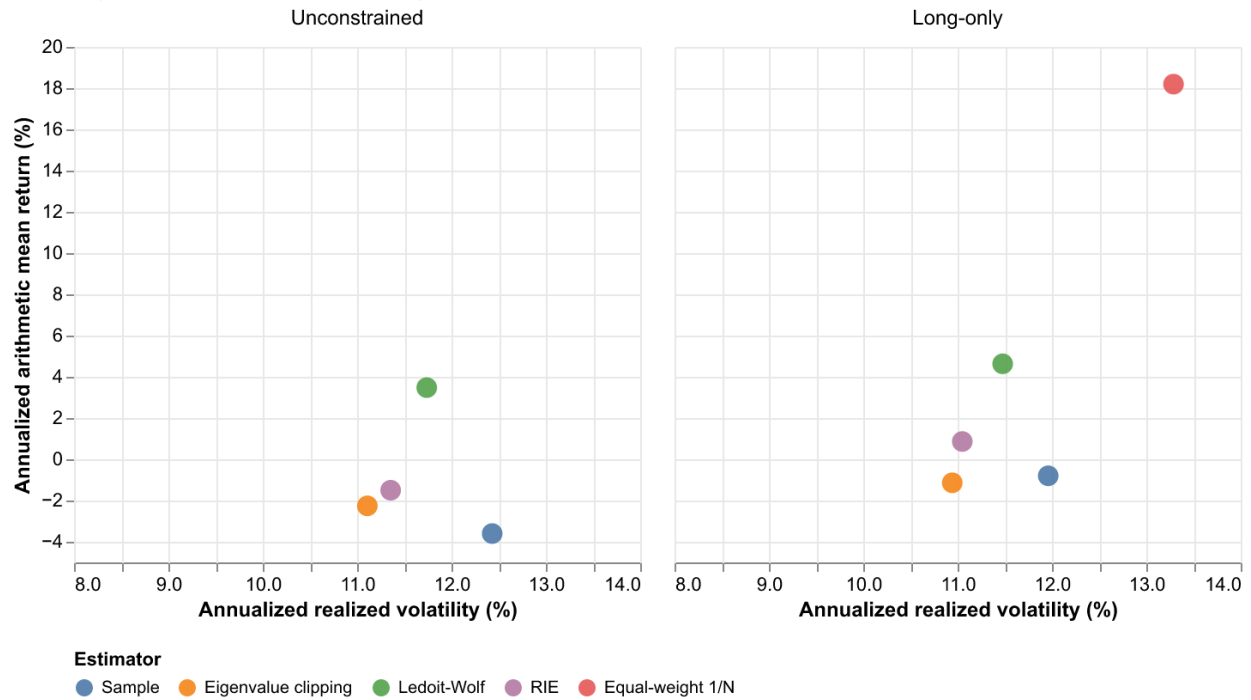


Figure 2: Annualized arithmetic return and volatility for the rolling 21-observation buy-and-hold portfolios. Both panels use a 504-observation lookback.

The equal-weight benchmark remains ahead under assumed trading costs

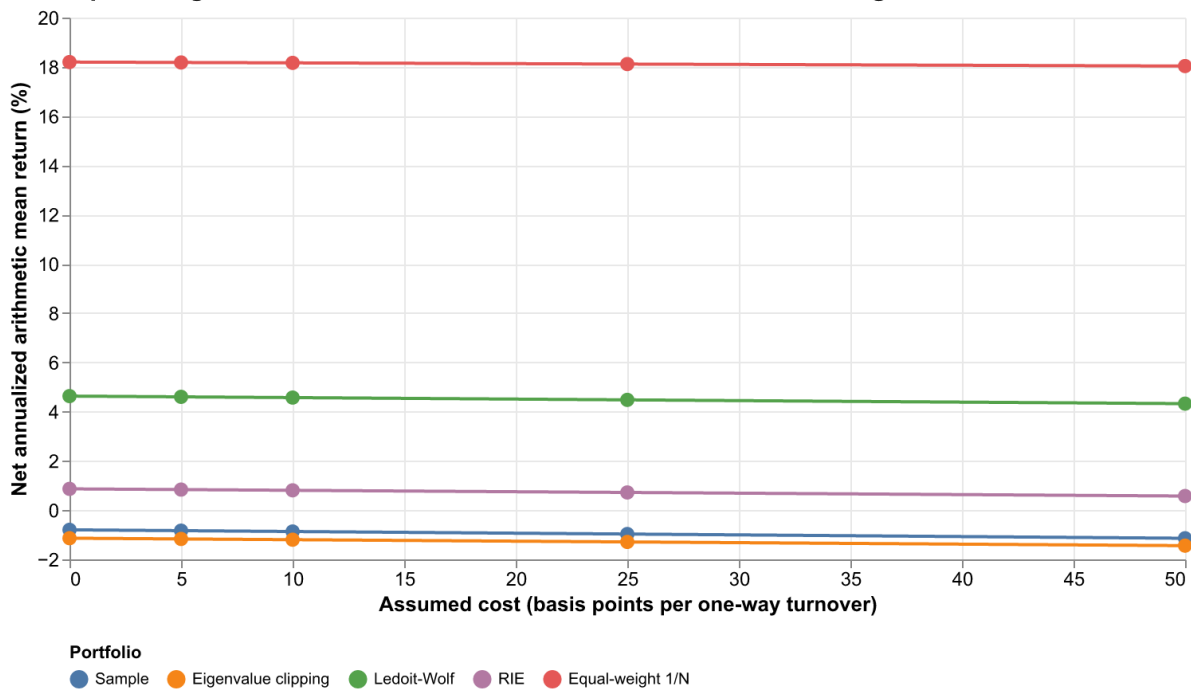


Figure 3: Long-only rolling portfolio returns under assumed proportional transaction-cost scenarios. The costs are hypothetical rather than measured Mexican trading costs.

reaches the same practical warning: point-in-time estimator rankings and implementation rules matter.

For emerging-market investment research, this distinction is material. Performance claims based only on estimated or realized minimum variance can omit the benchmark an allocator could actually hold and the turnover needed to maintain optimized weights. Our cost grid demonstrates how to report such sensitivities, but it does not establish local execution costs, investability, or a causal effect of market institutions. Those questions require quote-level spreads, trading volumes, market impact, a local risk-free series, and institutional portfolio constraints.

Several limitations remain substantive. The 37 names were mapped from a retrospective security list, not a point-in-time BMV/BIVA membership database; survivor and selection bias cannot be ruled out. One 20% holdout and one overlapping rolling sensitivity are not independent replications. Bootstrap and HAC inference condition on weights and the selected universe. The cost exercise uses assumed basis-point rates rather than observed execution prices and omits liquidity limits, market impact, short-sale frictions, taxes, and initial deployment. The risk-free rate is fixed at 8% rather than matched to a local yield series. These limitations prevent claims about an investable strategy, market-wide diversification, or generalization to other emerging economies.

5 Conclusion

For a daily gross-total-return panel of 37 Mexican equities, spectral covariance cleaning changes the estimated portfolio comparison but does not yield a consistent Sharpe advantage. RIE lowers fixed-holdout volatility relative to SM while its Sharpe is lower; under long-only constraints, $1/N$ has the highest Sharpe point estimate. The same benchmark remains ahead in the rolling analysis and across the stated assumed-cost scenarios. The evidence supports a practical conclusion for this sample: covariance cleaning should be assessed jointly with simple benchmarks, portfolio constraints, turnover, and the timing of rebalancing.

The next research step is a multi-market, point-in-time panel of emerging equities with delisted securities, consistent total-return histories, and institution-specific trading-cost data. Such a design could test whether the benchmark and estimator patterns documented here extend beyond this selected Mexican panel and whether they have implications for local investment institutions.

Data availability. The Bloomberg workbook containing daily TOT_RETURN_INDEX_GROSS_DVDS data is licensed and is not redistributed. The public repository (<https://github.com/IvliCaesar/bmv-spectral-portfolio-cleaning>) contains aggregate outcomes, a ticker-to-workbook mapping, analysis code, and figure-generation scripts; it contains no daily prices, returns, security weights, or licensed exports. Reproduction requires lawful access to a compatible Bloomberg export with the documented tabs and fields.

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